

# Causal Inference in Economics and Marketing

## Section 09: Synthetic Control

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September 06, 2026

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# 1 The problem

## 1.1 One treated unit, many donors, no parallel trends

Section 07 compared groups of units before and after a policy. Its assumption was parallel trends: without the policy, the treated group would have moved like the comparison group. Many policies do not come in groups. A country adopts a carbon tax. A state passes a tobacco law. A region suffers a terrorist conflict. One treated unit, observed over many periods, and a pool of untreated units that may or may not resemble it.

Sweden is the running example of this lecture (Andersson 2019). In 1990 Sweden extended its value-added tax to transport fuel and in 1991 added a carbon tax. The question is the effect on per-capita CO2 emissions from transport. The data cover Sweden and 14 OECD countries from 1960 to 2005: 30 pre-treatment years and 16 post-treatment years.

```
carbon <- as.data.frame(haven::read_dta(file.path(data_path, "carbontax_data.dta")))
carbon$treated <- as.integer(carbon$country == "Sweden" & carbon$year >= 1990)
donor_avg <- carbon %>% filter(country != "Sweden") %>% group_by(year) %>%
  summarise(y = mean(CO2_transport_capita), .groups = "drop") %>% mutate(series = "OECD donor average")
sweden <- carbon %>% filter(country == "Sweden") %>%
  transmute(year, y = CO2_transport_capita, series = "Sweden")

ggplot(bind_rows(sweden, donor_avg), aes(x = year, y = y, colour = series, linetype = series)) +
  geom_line() + geom_vline(xintercept = 1989.5, linetype = "dotted") +
  scale_colour_manual(values = c("Sweden" = "black", "OECD donor average" = "grey45")) +
  labs(x = NULL, y = "Metric tons per capita", colour = NULL, linetype = NULL) + sc_theme()
```

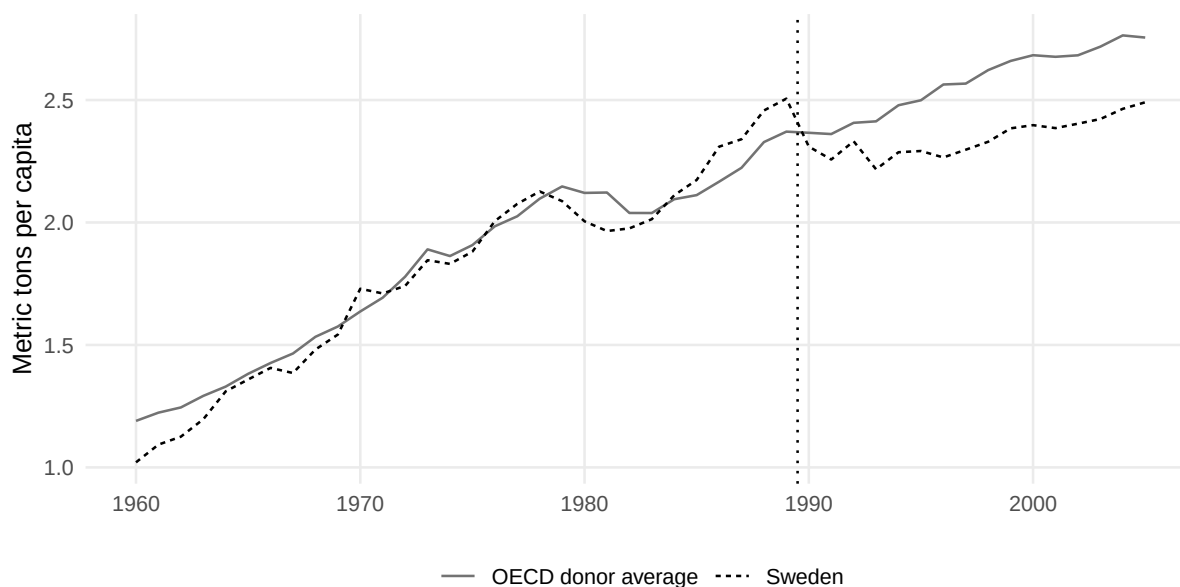


Figure 1: Per-capita CO2 emissions from transport, Sweden and the average of 14 OECD donor countries. The series diverge well before 1990.

DiD would compare Sweden with the donor average. Figure 1 shows why that fails. Sweden's emissions rise faster than the average through the 1960s and fall below it in the 1980s. Nothing in the pre-treatment data supports parallel trends. With one treated unit there is also no group of treated units to average over, and no cluster to compute a standard error from.

Synthetic control (Abadie and Gardeazabal 2003; Abadie, Diamond, and Hainmueller 2010) replaces the donor average by a weighted average. The weights are nonnegative, sum to one, and are chosen so that the weighted donors reproduce Sweden's pre-treatment path. The same weights then give the counterfactual after 1990. Two things change relative to DiD.

- The comparison unit is chosen by the data. Parallel trends is no longer assumed; it is built in the pre-treatment period and checked by eye.
- The model behind it allows unobserved common shocks with unit-specific loadings, not only additive unit and time effects. A donor mix that matches Sweden's pre-treatment path also matches its exposure to those shocks, if the fit is good enough.

```
sc_sweden <- synth_control(carbon, id = "country", time = "year", y = "CO2_transport_capita", d = "treated")
did_sweden <- sdid_weights(carbon, id = "country", time = "year", y = "CO2_transport_capita", d = "treated",
                           estimator = "did")

sc_sweden
#> Synthetic control: average post-treatment effect -0.2837
#> 1 treated and 14 donor units; 30 pre- and 16 post-treatment periods; 30 predictors (V: equal)
#> Pre-treatment RMSPE 0.0343; weight concentration ||w||2 = 0.252 (3.97 effective donors)
#> Weights above 0.001: Denmark (0.420), Belgium (0.202), New Zealand (0.136), United States (0.092), Greece (0.067), Spain (0.067)
c(did = did_sweden$estimate, synthetic_control = sc_sweden$estimate)
#> did synthetic_control
#> -0.2137205 -0.2836777

plot_synth(sc_sweden, type = "path")
```

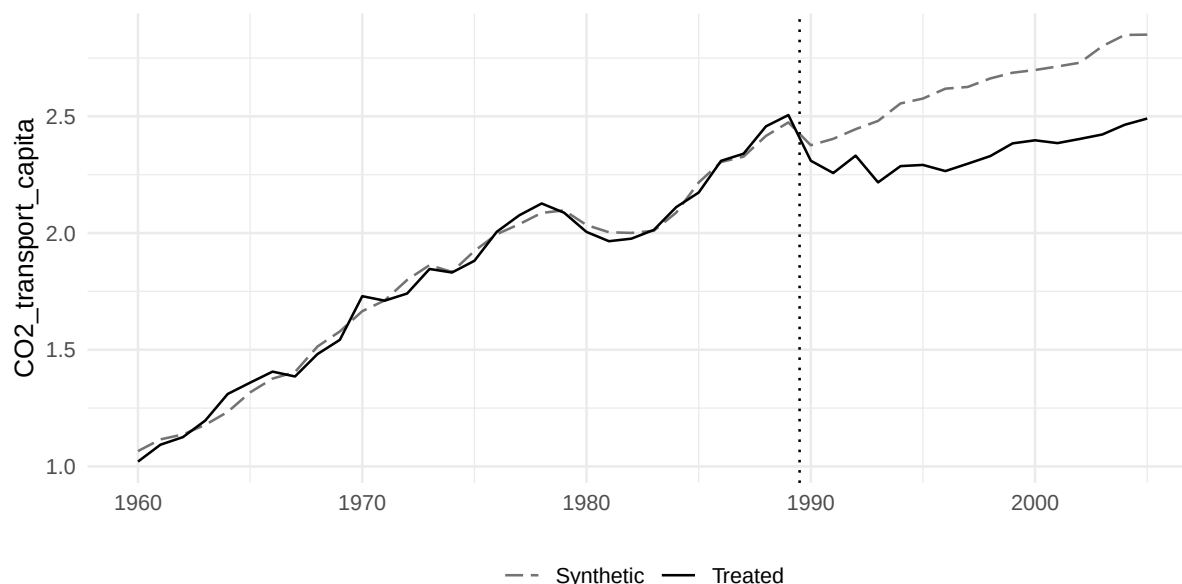


Figure 2: Sweden and its synthetic control built from all pre-treatment outcomes. The gap after 1990 is the estimated effect.

The synthetic Sweden tracks the real one closely before 1990 and diverges after. The average post-treatment gap is -0.28 tons per capita, a reduction of roughly ten percent. DiD gives a different number, and Figure 1 says which one to trust.

Three questions follow and organize the lecture. When do such weights recover the counterfactual, and when do they fail? How is uncertainty assessed with one treated unit? And how is an application built and defended in practice?

## 1.2 One experiment, six estimators

The natural model for this setting is a linear factor model (Section 2):

$$Y_{jt}^N = c_j + \delta_t + \lambda_t' \mu_j + \varepsilon_{jt}.$$

Units differ in their fixed effects  $c_j$  and in how they load ( $\mu_j$ ) on common factors  $\lambda_t$ . DiD assumes the loadings are equal. Synthetic control does not. The experiment below simulates this model with `sim_synth_panel()`: 20 donors, 20 pre-treatment periods, 5 post-treatment periods, two factors, a true effect of one. Two designs follow Ferman and Pinto (2021, Table 1).

- **Design A, level in the tail.** The treated unit's fixed effect sits two standard deviations above the donors'. Its loadings are ordinary. The factors have mean zero after treatment, so the DiD assumption holds.
- **Design B, loading in the tail with a break.** The treated unit loads 1.5 standard deviations above the donors on the first factor, and that factor shifts up by one standard deviation after treatment. Treatment is correlated with a time-varying unobservable.

| Estimator                   | Weights                                         | Intercept               | Post-treatment correction          |
|-----------------------------|-------------------------------------------------|-------------------------|------------------------------------|
| DiD                         | uniform                                         | unit and period effects | none                               |
| Synthetic control           | simplex, matched on outcome levels              | none                    | none                               |
| Demeaned synthetic control  | simplex, matched on outcome deviations          | yes                     | none                               |
| Augmented synthetic control | as demeaned                                     | yes                     | ridge regression on the predictors |
| Synthetic DiD               | simplex unit weights and period weights         | unit and period effects | none                               |
| Unconstrained regression    | least squares of the treated unit on the donors | yes                     | none                               |

```
results <- run_sc_experiment(n_sims = n_sims, seed = 1)
tab <- summarise_sc_experiment(results, effect = 1)
pick <- function(design_prefix, estimator, column) {
  tab[[column]][startsWith(tab$design, design_prefix) & tab$estimator == estimator]
}
```

```
plot_sc_experiment(results, effect = 1)
```

Table 2: Six estimators on the linear factor model, true effect 1, 200 simulations per design. Pre-RMSPE is the pre-treatment root mean squared prediction error; concentration is the squared norm of the donor weights.

| Design | Estimator                   | Bias  | SD    | RMSE  | Pre-RMSPE | Concentration |
|--------|-----------------------------|-------|-------|-------|-----------|---------------|
| A      | DiD                         | 0.076 | 1.297 | 1.296 | NA        | 0.050         |
| A      | Synthetic control           | 0.833 | 0.880 | 1.210 | 1.361     | 0.403         |
| A      | Demeaned synthetic control  | 0.016 | 0.782 | 0.781 | 0.923     | 0.294         |
| A      | Augmented synthetic control | 0.041 | 1.342 | 1.339 | 0.457     | 0.294         |
| A      | Synthetic DiD               | 0.041 | 0.723 | 0.722 | NA        | 0.089         |
| A      | Unconstrained regression    | 0.565 | 6.629 | 6.637 | 0.000     | 47.693        |
| B      | DiD                         | 1.540 | 1.397 | 2.077 | NA        | 0.050         |
| B      | Synthetic control           | 0.541 | 1.009 | 1.142 | 1.205     | 0.353         |
| B      | Demeaned synthetic control  | 0.470 | 0.790 | 0.918 | 1.037     | 0.334         |
| B      | Augmented synthetic control | 0.062 | 1.189 | 1.187 | 0.519     | 0.334         |
| B      | Synthetic DiD               | 0.274 | 0.837 | 0.878 | NA        | 0.107         |
| B      | Unconstrained regression    | 0.302 | 6.521 | 6.512 | 0.000     | 48.311        |

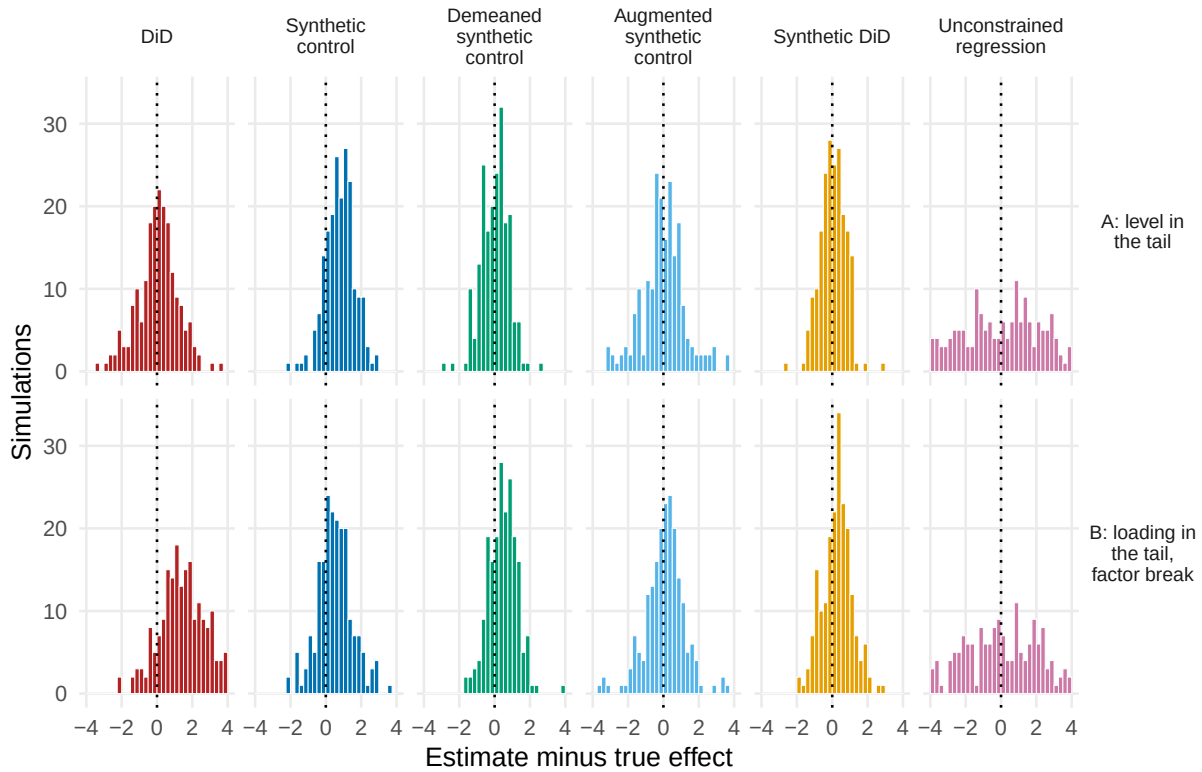


Figure 3: Sampling distributions of the six estimators minus the true effect, by design, truncated at plus or minus 4. Most draws of the unconstrained regression fall outside that range.

### 1.3 Reading the results

- **DiD** is unbiased in Design A (bias 0.08) and badly biased in Design B (1.54). The donor average loads less on the shifted factor than the treated unit, and the shift lands on the estimate. This is the failure that motivates the lecture: a comparison group with the wrong exposure to common shocks.

- **Synthetic control** is biased in Design A (0.83). Weights on the simplex can only reach levels inside the donors' range; a treated unit above that range cannot be matched, and the level gap becomes a bias. Its pre-treatment fit is visibly worse (pre-RMSPE 1.36). In Design B it reduces the bias of DiD to 0.54 but does not remove it.
- **Demeaned synthetic control** matches deviations from each unit's own mean. It has no level problem in Design A (bias 0.02), and a smaller standard deviation than DiD (0.78 against 1.30). In Design B it is biased, like every estimator, because noise keeps the weights from fully matching the loadings; the bias (0.47) is a fraction of DiD's.
- **Augmented synthetic control** adds a regression correction for the remaining imbalance in the predictors. It trades bias for variance (Design B bias 0.06, SD 1.19). Its pre-treatment fit is better by construction, because the correction is fitted on those periods; that number is not a diagnostic.
- **Synthetic DiD** combines unit weights, period weights, and two-way effects (Section 07). It performs like the demeaned synthetic control in Design A and somewhat better in Design B.
- **Unconstrained regression** of the treated unit on all donors fits the pre-treatment period perfectly, because there are as many donors as pre-treatment periods, and then extrapolates noise. Its RMSE is the largest in both designs. A perfect pre-treatment fit is not evidence of a good counterfactual when the donors outnumber the periods.

Three lessons carry through the lecture. Matching levels and matching trends are different things, and the intercept decides which one is asked. A good pre-treatment fit is necessary but not sufficient: it can come from overfitting, and it does not protect against a shock that treated and donors feel differently. And weight concentration is a diagnostic: among the simplex estimators, the ones that fail most spread their weights least.

## 1.4 When this is the right tool

Synthetic control is the design for a treated aggregate unit with a long pre-treatment history and a pool of untreated units that are exposed to the same common shocks. Its requirements, in the order they are usually violated:

- **Unaffected donors.** No spillover from the treated unit to the donors, and no donor with its own policy change in the window.
- **No anticipation.** The pre-treatment path must be a path without treatment; date the treatment at its announcement if units respond early.
- **A long pre-treatment period relative to the donor pool.** With few periods and many donors, the fit is achieved by chance.
- **A treated unit inside the donors' range.** Weights on the simplex interpolate; a treated unit at the extreme of the distribution cannot be reproduced without extrapolation.
- **Stable factor structure.** The relation between the treated unit and its synthetic control, learned before treatment, must persist after it.

When these fail, the alternatives are DiD with a justified comparison group, synthetic DiD, or a design that does not rely on the donors.

**Roadmap.** Section 2 states the factor model, shows why the weights work under a perfect fit, and what breaks under an imperfect fit. Section 3 builds the synthetic control: predictors, the importance weights, constraints, and augmentation. Section 4 covers inference with one treated unit: placebos, conformal tests, and the specification test. Section 5 collects the robustness checks and the checklist of Ferman and Pinto. Section 6 turns to the applications and Section 7 to the recent literature. Proofs are sketched in the text and completed in the Appendix.

## 2 Just enough framework

### 2.1 Setup and estimand

Units  $j = 0, 1, \dots, J$  are observed in periods  $t = 1, \dots, T$ . Unit 0 is treated from period  $T_0 + 1$  on; units  $1, \dots, J$  are the donors. Potential outcomes are  $Y_{jt}^N$  without treatment and  $Y_{jt}^I = Y_{jt}^N + \alpha_{jt}$  with it. The observed outcome is  $Y_{jt} = Y_{jt}^N + D_{jt} \alpha_{jt}$  with  $D_{jt} = 1\{j = 0, t > T_0\}$ .

The estimand is the effect on the treated unit in each post-treatment period,  $\alpha_{0t} = Y_{0t} - Y_{0t}^N$  for  $t > T_0$ , and its average over the post-treatment periods. It is not an average over units. Only  $Y_{0t}^N$  is missing.

Two assumptions come before any model.

- **No anticipation.**  $Y_{0t} = Y_{0t}^N$  for  $t \leq T_0$ . The pre-treatment path is a path without treatment.
- **No interference.**  $Y_{jt} = Y_{jt}^N$  for every donor and every  $t$ . Donors are unaffected by the treated unit's policy.

Both are restrictions on the donor pool and the treatment date, not on the estimator.

**Definition (synthetic control).** For weights  $w$  in the simplex  $\Delta = \{w \in \mathbb{R}^J : w_j \geq 0, \sum_j w_j = 1\}$ ,

$$\hat{Y}_{0t}^N(w) = \sum_{j=1}^J w_j Y_{jt}, \quad \hat{\alpha}_{0t}(w) = Y_{0t} - \sum_{j=1}^J w_j Y_{jt}.$$

The synthetic control weights minimize the pre-treatment discrepancy,

$$\hat{w} = \arg \min_{w \in \Delta} \sum_{t \leq T_0} \left( Y_{0t} - \sum_j w_j Y_{jt} \right)^2$$

when every pre-treatment outcome is a predictor, or  $\|Z_0 - Zw\|_V^2 = (Z_0 - Zw)'V(Z_0 - Zw)$  for a vector of predictors  $Z_0$  (outcome lags and covariates) and importance weights  $V$ .

### 2.2 The linear factor model

The theory of the estimator is stated for

$$Y_{jt}^N = c_j + \delta_t + \lambda_t' \mu_j + \varepsilon_{jt},$$

with unit effects  $c_j$ , period effects  $\delta_t$ ,  $F$  common factors  $\lambda_t$  with unit-specific loadings  $\mu_j$ , and idiosyncratic shocks  $\varepsilon_{jt}$  with  $\mathbb{E}[\varepsilon_{jt}] = 0$ . Two-way fixed effects is the case  $\mu_j \equiv \mu$ : the factors then act like period effects. The factor model lets a common shock hit units with different intensity, which is the case parallel trends rules out.

Treatment assignment is treated as fixed, together with  $(c_j, \mu_j)$ . The identifying condition (Ferman and Pinto 2021, Assumption 3) is that the idiosyncratic shocks have mean zero given the assignment. Treatment may depend on the treated unit's level  $c_0$  and loadings  $\mu_0$ ; it may not depend on shocks specific to the treated unit. This is what "selection on unobservables" means here: unobserved confounders are allowed as long as they also move the donors.

|                | DiD                                                                                      | Synthetic control                                                                                |
|----------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Counterfactual | donor average, shifted by the pre-treatment gap between the treated unit and the average | weighted donors, weights fitted before treatment                                                 |
| Absorbs        | $c_j$ and $\delta_t$                                                                     | $\delta_t$ (weights sum to one), plus $c_0$ and $\lambda_t' \mu_0$ if the weights reproduce them |

|       | DiD                                                                  | Synthetic control                                                                     |
|-------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Needs | $\mu_0 = \bar{\mu}$ , or $\mathbb{E}[\lambda_t] = 0$ after treatment | weights $w$ with $c'w = c_0$ and $\mu'w = \mu_0$ , and a fit good enough to find them |

## 2.3 Why the weights work: skeleton of the argument

Substitute the model into the estimator. For any  $w \in \Delta$  and any  $t$ ,

$$\hat{\alpha}_{0t}(w) - \alpha_{0t} = \underbrace{(c_0 - c'w)}_{\text{level mismatch}} + \underbrace{\lambda'_t(\mu_0 - \mu'w)}_{\text{factor mismatch}} + \underbrace{(\varepsilon_{0t} - \varepsilon'_t w)}_{\text{noise}},$$

where  $c = (c_1, \dots, c_J)'$ ,  $\mu = (\mu_1, \dots, \mu_J)'$ , and  $\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Jt})'$ . The period effect  $\delta_t$  cancels because the weights sum to one. The noise term has mean zero. Everything depends on whether the fitted weights make the first two terms small.

**Step 1: perfect fit (Abadie, Diamond, and Hainmueller 2010).** Suppose weights  $w^*$  exist with  $c'w^* = c_0$  and  $\mu'w^* = \mu_0$ , and suppose the fitted weights reproduce the treated unit exactly over the pre-treatment period. Then the pre-treatment gap is the noise term evaluated at  $\hat{w}$ , and a mismatch in the loadings would have to be offset by the noise in every one of  $T_0$  periods. The bias of  $\hat{\alpha}_{0t}$  is bounded by a term that shrinks as  $T_0$  grows, with a rate governed by the noise variance and the number of donors. Appendix 8.1 states the bound. The lesson is practical: a long pre-treatment period with a close fit is the case the method was designed for.

**Step 2: imperfect fit (Ferman and Pinto 2021).** Drop the perfect-fit condition. With every pre-treatment outcome as a predictor, the objective divided by  $T_0$  converges, as  $T_0 \rightarrow \infty$  with  $J$  fixed, to

$$Q_0(w) = (c_0 - c'w)^2 + (\mu_0 - \mu'w)' \Omega_0 (\mu_0 - \mu'w) + \sigma^2 (1 + w'w),$$

where  $\Omega_0$  is the long-run second moment of the factors and  $\sigma^2$  the noise variance (Appendix 8.2). The fitted weights converge to  $\bar{w} = \arg \min_{\Delta} Q_0$ . The last term is the variance of the weighted noise, and it rewards spreading the weights. Unless  $\sigma^2 = 0$ , the minimizer does not set the first two terms to zero, even when weights that would do so exist. So

$$\hat{\alpha}_{0t} \rightarrow_p \alpha_{0t} + (c_0 - c'\bar{w}) + \lambda'_t(\mu_0 - \mu'\bar{w}) + (\varepsilon_{0t} - \varepsilon'_t \bar{w}).$$

Four consequences, each visible in Table 2.

- **Level bias.** If  $\bar{w}$  misses  $c_0$ , the estimator is biased even when the factors have mean zero after treatment. This is Design A, and it is a bias DiD does not have.
- **The demeaned fix.** Subtracting each unit's pre-treatment mean removes the level term from  $Q_0$ . The demeaned synthetic control is then unbiased under exactly the DiD condition,  $\mathbb{E}[\lambda_t] = 0$  for  $t > T_0$ , and has weakly smaller asymptotic variance than DiD when the factor and noise moments are stable (Ferman and Pinto, Propositions 2 and 3). The intercept costs nothing in identification and buys efficiency.
- **Time-varying confounders.** If  $\mathbb{E}[\lambda_t] \neq 0$  after treatment, both are biased: DiD by  $\bar{\lambda}'(\mu_0 - \bar{\mu})$  and the demeaned synthetic control by  $\bar{\lambda}'(\mu_0 - \mu'\bar{w})$ . The second is usually smaller because  $\bar{w}$  is pulled toward matching the loadings, but it is not zero. This is Design B.
- **Concentration as a warning.** The trade-off in  $Q_0$  is between matching and spreading. Weights that are concentrated on few donors signal that matching was hard, and the simulations of Ferman and Pinto show the bias rising with  $\|w\|^2$ .

The following check makes Step 2 concrete. The treated unit's level and loadings are an exact convex combination of three donors' (weights 0.5, 0.3, 0.2), so  $w^*$  exists, and every unit has noise of standard deviation one. The table reports the distance between the fitted weights and  $w^*$ , the loading mismatch  $\|\mu_0 - \mu'\hat{w}\|$ , and the concentration  $\|\hat{w}\|^2$  as  $T_0$  grows.



```

w_true <- c(0.5, 0.3, 0.2, rep(0, 5))
limit_check <- function(t_pre) {
  dat <- simulate_convex_treated(n_donors = 8, t_pre = t_pre, t_post = 5, sd_e = 1, w_true = w_true, seed = 1)
  fit <- synth_control(dat, id = "id", time = "time", y = "y", d = "d")
  mu <- attr(dat, "donor_loadings")
  c(t_pre = t_pre, distance_to_w_star = sqrt(sum((fit$weights$weight - w_true)^2)),
    loading_mismatch = sqrt(sum((crossprod(mu, fit$weights$weight - w_true)^2)),
    concentration = fit$concentration[["12"]])
}
round(rbind(limit_check(50), limit_check(500), limit_check(5000)), 3)
#>      t_pre distance_to_w_star loading_mismatch concentration
#> [1,]    50          0.587          0.164          0.278
#> [2,]   500          0.412          0.124          0.187
#> [3,]  5000          0.343          0.141          0.224

```

The distance to  $w^*$  does not vanish with  $T_0$ . The weights settle at a more diluted vector than  $w^*$  (whose concentration is 0.38), and the loading mismatch that comes with it is permanent. That mismatch is harmless while the factors average zero, and becomes a bias when they do not.

## 2.4 Weights plus an outcome model

Every estimator of Section 1 has the form

$$\hat{Y}_{0t}^N = \hat{m}_t(Z_0) + \sum_j w_j(Y_{jt} - \hat{m}_t(Z_j)),$$

an outcome model evaluated at the treated unit plus a weighted average of the donors' residuals from that model (Ben-Michael, Feller, and Rothstein 2021; Doudchenko and Imbens 2016).

| Estimator                   | Outcome model $\hat{m}_t$                      | Weights                        |
|-----------------------------|------------------------------------------------|--------------------------------|
| DiD                         | unit and period effects                        | uniform                        |
| Synthetic control           | none                                           | simplex, on levels             |
| Demeaned synthetic control  | unit effect                                    | simplex, on deviations         |
| Augmented synthetic control | ridge regression of $Y_{jt}$ on the predictors | simplex, on deviations         |
| Synthetic DiD               | unit and period effects                        | simplex units, simplex periods |
| Unconstrained regression    | none                                           | least squares                  |

The error of the general form is

$$\hat{Y}_{0t}^N - Y_{0t}^N = (\hat{m}_t(Z_0) - m_t(Z_0)) - \sum_j w_j(\hat{m}_t(Z_j) - m_t(Z_j)) + \left(\sum_j w_j m_t(Z_j) - m_t(Z_0)\right) + \text{noise},$$

with  $m_t$  the true conditional mean. If the weights balance the predictors exactly and  $m_t$  is linear in them, the third term is zero whatever the model. If the model is right, the first two terms are zero whatever the weights. Otherwise the leading error is the product of the model error and the imbalance. This is the structure of Section 04: weights play the role of the propensity score, the outcome model plays the role of the regression, and the remainder is a product. The augmented synthetic control is the doubly robust member of the family, which is why it removed most of the bias in Design B. The price is variance, and it is paid through extrapolation: the outcome model predicts outside the donors' range.

## 3 Building the synthetic control

Each subsection follows the same order: the choice, what it does to the weights, the code on Sweden, and when it breaks.

Table 5: Donor weights for Sweden under three predictor choices. Andersson (2019, Table 2) reports Denmark 0.384, Belgium 0.195, New Zealand 0.177, Greece 0.090, United States 0.088, Switzerland 0.061.

| country       | all_lags | andersson_equal_v | andersson_mspe_v |
|---------------|----------|-------------------|------------------|
| Denmark       | 0.420    | 0.000             | 0.387            |
| New Zealand   | 0.136    | 0.055             | 0.217            |
| Belgium       | 0.202    | 0.479             | 0.167            |
| Greece        | 0.067    | 0.149             | 0.116            |
| United States | 0.092    | 0.148             | 0.087            |
| Switzerland   | 0.013    | 0.132             | 0.027            |
| Iceland       | 0.021    | 0.000             | 0.000            |
| Spain         | 0.047    | 0.000             | 0.000            |

### 3.1 Predictors and importance weights

**The choice.** Two conventions define the predictors  $Z$ . The first uses every pre-treatment outcome, so the weights reproduce the whole path. The second uses a short list of covariates averaged over a window, plus a few outcome lags, each weighted by an importance weight in  $V$ . Andersson (2019) uses GDP per capita, vehicles per capita, gasoline consumption per capita, and urban population, averaged over 1980 to 1989, plus emissions in 1989, 1980, and 1970.

**What it does.** With covariates,  $V$  decides which predictors the weights must match. **Synth** chooses  $V$  by a nested problem: for each  $V$ , solve the inner simplex problem for  $w(V)$ ; then choose  $V$  to minimize the pre-treatment mean squared prediction error of the outcome path. Kaul et al. (2022) show that when every outcome lag is a predictor, the outer problem gives the covariates zero weight: the lags alone determine  $w$ . Covariates matter only when the lags are few.

```
xs <- c("GDP_per_capita", "vehicles_capita", "gas_cons_capita", "urban_pop")
set.seed(1)
sc_and <- synth_control(carbon, id = "country", time = "year", y = "CO2_transport_capita", d = "treated",
  x = xs, lags = c(1989, 1980, 1970), pre_window = 1980:1989, v = "mspe",
  v_control = list(maxit = 2000, starts = 3))
sc_and_eq <- synth_control(carbon, id = "country", time = "year", y = "CO2_transport_capita", d = "treated",
  x = xs, lags = c(1989, 1980, 1970), pre_window = 1980:1989, v = "equal")
weights_tab <- Reduce(function(a, b) merge(a, b, by = "country", all = TRUE), list(
  setNames(sc_sweden$weights, c("country", "all_lags")),
  setNames(sc_and_eq$weights, c("country", "andersson_equal_v")),
  setNames(sc_and$weights, c("country", "andersson_mspe_v"))
))
weights_tab <- weights_tab[order(-weights_tab$andersson_mspe_v), ]
knitr::kable(weights_tab[weights_tab$andersson_mspe_v > 0.001 | weights_tab$all_lags > 0.001, ], digits = 3,
  row.names = FALSE, booktabs = TRUE,
  caption = "Donor weights for Sweden under three predictor choices. Andersson (2019, Table 2) reports Denmark 0.384,

knitr::kable(sc_and$balance[, c("predictor", "treated", "synthetic", "donor_mean", "v")], digits = 3,
  row.names = FALSE, booktabs = TRUE,
  col.names = c("Predictor", "Sweden", "Synthetic Sweden", "Donor mean", "V"),
  caption = "Predictor balance for Andersson's specification (his Table 1). The donor mean is unweighted.")
```

Three readings. The all-lags weights and Andersson's weights agree on the main donors (Denmark, Belgium, New Zealand, the United States) and differ in the tail. The  $V$  weights of the nested problem concentrate on two predictors; the pre-treatment MSPE is flat in  $V$  near its minimum, so different optimizers land on different  $V$  with almost the same  $w$ , and **Synth**'s  $V$  is not the one above.  $V$  is a means to an end: report it, but judge the fit and the weights. The balance table is the object to read. Sweden and synthetic Sweden agree on every predictor except gasoline consumption, exactly as in the paper.

**When it breaks.** Few lags and many covariates invite a fit driven by the covariates rather than the path. Many lags with a short pre-period invite overfitting (Section 3.3). Predictors measured after treatment are

Table 6: Predictor balance for Andersson’s specification (his Table 1). The donor mean is unweighted.

| Predictor                  | Sweden    | Synthetic Sweden | Donor mean | V     |
|----------------------------|-----------|------------------|------------|-------|
| GDP_per_capita             | 20121.479 | 19531.166        | 18706.663  | 0.000 |
| vehicles_capita            | 405.562   | 405.563          | 402.368    | 0.284 |
| gas_cons_capita            | 456.178   | 404.386          | 425.510    | 0.000 |
| urban_pop                  | 83.099    | 83.099           | 75.238     | 0.714 |
| CO2_transport_capita(1989) | 2.506     | 2.482            | 2.371      | 0.001 |
| CO2_transport_capita(1980) | 2.005     | 2.033            | 2.121      | 0.001 |
| CO2_transport_capita(1970) | 1.729     | 1.667            | 1.637      | 0.000 |

never allowed.

### 3.2 Constraints and extrapolation

**The choice.** Weights on the simplex interpolate: the synthetic unit lies inside the convex hull of the donors, predictor by predictor. Adding an intercept (`demean = TRUE`) lets the level differ while the trends match. Dropping the sum-to-one constraint or the sign constraint (Doudchenko and Imbens 2016; Hsiao, Ching, and Wan 2012) allows extrapolation in every direction.

```
fits <- list(
  simplex = sc_sweden,
  demeaned = synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", demean = TRUE),
  nonnegative = synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", constraints = "nonnegative"),
  unconstrained = synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", constraints = "none")
)
t(sapply(fits, function(f) c(estimate = f$estimate, pre_rmspe = f$pre_rmspe, sum_w = sum(f$weights$weight),
  positive_weights = sum(f$weights$weight > 0.001), concentration = f$concentration[["l2"]]))))
#>           estimate pre_rmspe sum_w positive_weights concentration
#> simplex      -0.2836777 0.03430629 1.000000           8      0.2518521
#> demeaned     -0.2914985 0.03417556 1.000000           8      0.2647790
#> nonnegative  -0.3582306 0.03242864 1.285178           8      0.3567614
#> unconstrained -0.2927512 0.02825092 1.349599          10      1.1909378
```

**What it does.** The simplex gives a sparse, interpretable set of donors. The intercept changes the estimate little here because Sweden’s level is inside the donors’ range; it would matter for a country at the top or bottom of the distribution (Design A). The unconstrained regression has 30 pre-treatment periods and 14 donors, so it does not interpolate the data, but its weights are dense and of both signs, and its counterfactual is an extrapolation no one can defend donor by donor.

The range check in the balance table is the first guard against extrapolation: every predictor of the treated unit should lie within the donors’ range.

```
sc_and$balance[, c("predictor", "treated", "in_range")]
#>           predictor treated in_range
#> 1 GDP_per_capita 20121.479093      TRUE
#> 2 vehicles_capita 405.561575      TRUE
#> 3 gas_cons_capita 456.178040      TRUE
#> 4 urban_pop      83.098700      TRUE
#> 5 CO2_transport_capita(1989) 2.505603      TRUE
#> 6 CO2_transport_capita(1980) 2.004685      TRUE
#> 7 CO2_transport_capita(1970) 1.729497      TRUE
```

**When it breaks.** A treated unit at the extreme of the predictor distribution cannot be reproduced on the simplex, and the fit will show it. The honest responses are the intercept (which extrapolates the level only) or a different design; unconstrained weights hide the problem rather than solve it.

Table 7: Demeaned synthetic control with no treatment effect, 10 pre-treatment periods, and noise standard deviation 2: fit and post-treatment error by donor-pool size, 100 simulations each.

| Donors | Pre-RMSPE | Post-RMSPE | Bias   | SD    | RMSE  | Concentration |
|--------|-----------|------------|--------|-------|-------|---------------|
| 5      | 1.908     | 2.719      | -0.036 | 1.829 | 1.820 | 0.466         |
| 10     | 1.462     | 2.602      | -0.007 | 1.578 | 1.570 | 0.375         |
| 20     | 1.296     | 2.580      | -0.205 | 1.426 | 1.434 | 0.323         |
| 40     | 0.958     | 2.335      | -0.174 | 1.338 | 1.342 | 0.291         |
| 80     | 0.805     | 2.251      | 0.170  | 1.329 | 1.333 | 0.251         |

### 3.3 Pre-treatment fit and overfitting

**The choice.** Every synthetic control is judged first by its pre-treatment fit. The root mean squared prediction error over the pre-treatment periods (`pre_rmspe`) is the number; the path plot is the picture. Ferman and Pinto (2021) recommend a second picture: subtract the period mean of all units from every series and look at the fit again. Common trends that all units share are removed, and what remains is the part of the fit that the weights had to earn.

```
plot_synth(sc_sweden, type = "demeaned_path")
```

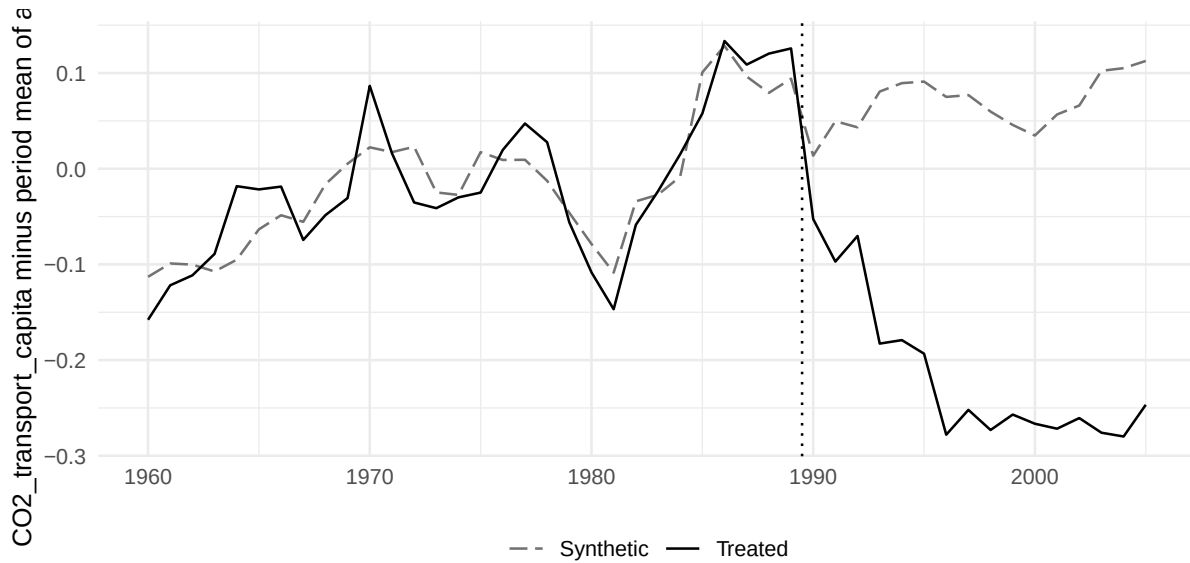


Figure 4: Sweden and its synthetic control after subtracting the period mean of all 15 countries. The fit before 1990 is what the weights achieved beyond the common trend.

**Overfitting.** A close fit is easy to obtain when the donors outnumber the pre-treatment periods, because the weights can absorb noise. The experiment below draws noisy panels with no treatment effect, 10 pre-treatment periods, and 5 to 80 donors. The pre-treatment fit improves with the donor pool while the post-treatment error does not.

```
overfit <- run_overfit_experiment(n_sims = 100, n_donors = c(5, 10, 20, 40, 80), t_pre = 10, sd_e = 2, seed = 2)
overfit_tab <- summarise_overfit(overfit)
```

With 80 donors the pre-treatment fit is 0.42 times that with 5 donors, and the post-treatment error is 0.83 times. The improvement in the fit is fitted noise. Read the pre-treatment fit together with the ratio of donors to periods, and prefer the post/pre comparison of the placebo test (Section 4.2) to the pre-treatment fit alone.

Table 8: Ridge augmentation on Sweden under three predictor sets. The correction is averaged over the post-treatment periods.

| Predictors               | Demeaned SC | Augmented | Lambda | Mean correction | Share of estimate |
|--------------------------|-------------|-----------|--------|-----------------|-------------------|
| all 30 lags              | -0.291      | -0.171    | 0.001  | -0.120          | 0.702             |
| Andersson's 7 predictors | -0.197      | -0.246    | 0.014  | 0.049           | -0.198            |
| all 30 lags, lambda = 1  | -0.291      | -0.301    | 1.000  | 0.009           | -0.031            |

**When it breaks.** Ferman and Pinto's Remark 3: a good fit with  $J$  large relative to  $T_0$  is not evidence that the loadings are matched. Sweden has 14 donors and 30 pre-treatment years, which is comfortable.

### 3.4 Regularization and augmentation

**The choice.** Two additions borrow from Section 07 and Section 04. A ridge penalty on the weights (**zeta**, as in synthetic DiD) spreads them across donors at the cost of fit. The ridge-augmented estimator (**augment** = "ridge") keeps the simplex weights and adds the correction  $\hat{m}_t(Z_0) - \sum_j w_j \hat{m}_t(Z_j)$  of Section 2.4, with  $\hat{m}_t$  a ridge regression of each period's outcome on the predictors, fitted on donors, and its penalty chosen by leave-one-donor-out cross-validation.

```
sc_dm <- fits$demeaned
aug_row <- function(label, ...) {
  base <- synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", demean = TRUE, ...)
  aug <- synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", demean = TRUE, augment = "ridge", ...)
  corr <- aug$augmentation$correction[aug$effects$post]
  data.frame(predictors = label, demeaned_sc = base$estimate, augmented = aug$estimate, lambda = aug$lambda,
             mean_correction = mean(corr), correction_share = mean(corr) / aug$estimate)
}
aug_tab <- rbind(
  aug_row("all 30 lags"),
  aug_row("Andersson's 7 predictors", x = xs, lags = c(1989, 1980, 1970), pre_window = 1980:1989),
  aug_row("all 30 lags, lambda = 1", lambda = 1)
)
knitr::kable(aug_tab, digits = 3, row.names = FALSE, booktabs = TRUE,
             col.names = c("Predictors", "Demeaned SC", "Augmented", "Lambda", "Mean correction", "Share of estimate"),
             caption = "Ridge augmentation on Sweden under three predictor sets. The correction is averaged over the post-treatment periods.")
```

**What it does.** The correction is the augmented estimator's own diagnostic. With all 30 lags as predictors and 14 donors, the ridge model has more parameters than observations; cross-validation keeps the penalty near zero and the correction is 70 percent of the estimate. With Andersson's seven predictors it is a fifth of the estimate, with the opposite sign. Neither is small. A correction comparable to the estimate says that the answer comes from the outcome model, not from the donors. In that case report the synthetic control estimate as the headline and the augmented one as a sensitivity check, or fix a larger penalty, which shrinks the augmented estimator back to the synthetic control (last row). The augmented estimator earns its keep when the pre-treatment fit is poor and the predictor set is small relative to the donor pool.

**When it breaks.** The ridge model is linear in the predictors. When the post-treatment outcomes depend on the pre-treatment path in a different way for the treated unit than for the donors, the correction adds bias rather than removing it. The Monte Carlo of Section 1 showed the other cost: variance.

## 4 Inference

### 4.1 What is random

There is one treated unit. The estimate is one gap per post-treatment period, and nothing is averaged across treated units. Two things are uncertain: the weights, which were estimated from noisy pre-treatment data, and the idiosyncratic shocks after treatment, which the weighted donors do not reproduce. A standard error

in the usual sense needs a population that the treated unit was sampled from, and there is none.

The literature answers with Fisher’s logic. Under a sharp null of no effect, the treated unit is exchangeable with the donors, or the post-treatment periods are exchangeable with the pre-treatment ones, and either exchangeability gives a reference distribution. Three tools follow.

- **In-space placebos** permute the treatment across units (Abadie, Diamond, and Hainmueller 2010).
- **In-time placebos and conformal inference** permute across periods (Abadie, Diamond, and Hainmueller 2015; Chernozhukov, Wuthrich, and Zhu 2021).
- **The specification test** of Ferman and Pinto (2021) permutes across periods to compare two estimators rather than to test an effect.

## 4.2 In-space placebos

**Procedure.** Reassign the treatment to each donor in turn, fit its synthetic control from the remaining donors with the same predictors, and record its gap path. The treated unit’s gap is compared with the distribution of placebo gaps. The statistic that adjusts for fit is the ratio of post-treatment to pre-treatment mean squared prediction error: a large post-treatment gap counts for more when the pre-treatment fit was tight. The p-value is the rank of the treated unit’s ratio among all units, divided by their number. Placebos that could not be fitted are usually dropped: Andersson keeps donors whose pre-treatment MSPE is at most 20 times Sweden’s.

```
pl_cut <- synth_placebo(sc_and, mspe_limit = 20)
pl_cut
#> In-space placebo test: 14 placebo units, 9 kept (pre-MSPE within 20x the treated unit's)
#> Treated post/pre MSPE ratio: 80.8; rank 1 of 10; p-value 0.1
head(pl_cut$mspe[order(pl_cut$mspe$rank), c("country", "pre_mspe", "post_mspe", "ratio", "rank")], 5)
#>      country  pre_mspe post_mspe  ratio rank
#> 1      Sweden 0.001205091 0.09737288 80.801292 1
#> 13     Spain 0.004158184 0.05055825 12.158732 2
#> 3      Belgium 0.003409653 0.03675327 10.779184 3
#> 14 Switzerland 0.006187207 0.04914881 7.943617 4
#> 9       Japan 0.004042379 0.02355518 5.827059 5
pl_lags <- synth_placebo(sc_sweden, mspe_limit = 20)
c(rank_andersson_predictors = pl_cut$mspe$rank[1], rank_all_lags = pl_lags$mspe$rank[1],
  p_andersson_predictors = pl_cut$p_value, p_all_lags = pl_lags$p_value)
#> rank_andersson_predictors      rank_all_lags      p_andersson_predictors
#>      1.0000000      2.0000000      0.1000000
#>      p_all_lags
#>      0.1818182

plot_synth(pl_cut)
```

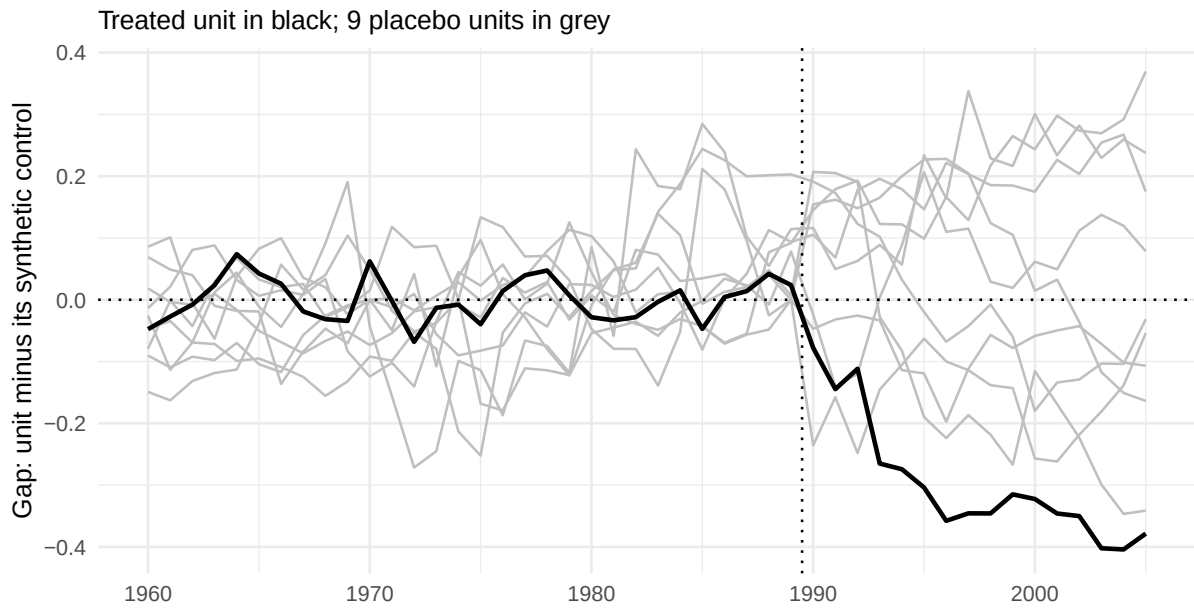


Figure 5: Gaps for Sweden (black) and for each donor treated as a placebo (grey). Sweden's post-1990 gap is the largest and most persistent.

```
plot_synth(pl_cut, type = "ratio")
```

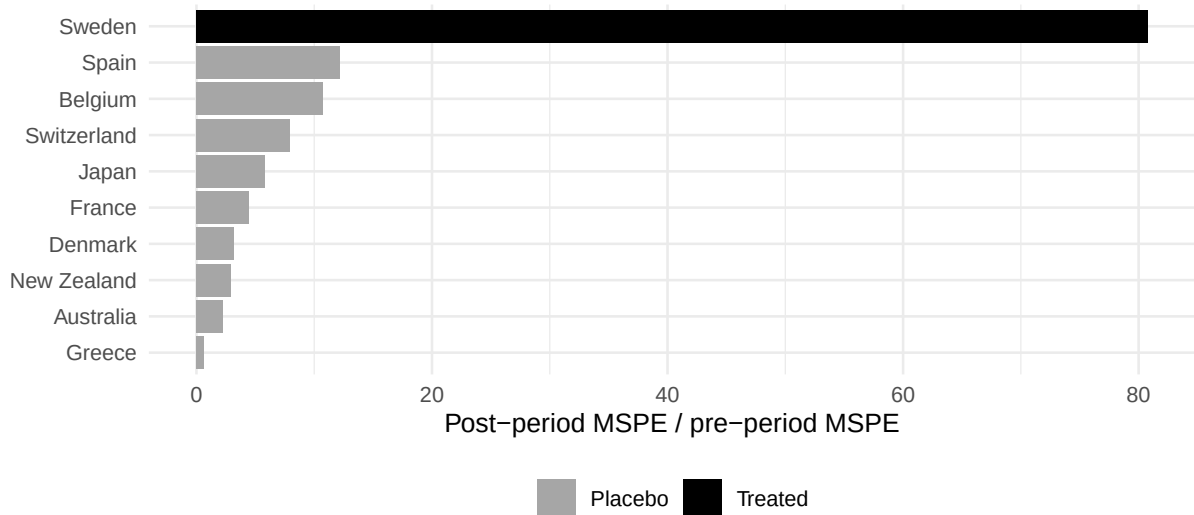


Figure 6: Ratio of post-treatment to pre-treatment mean squared prediction error, Sweden against the placebos kept after the 20-times cut.

**Reading it.** With Andersson's predictors, Sweden's ratio ranks first among the 10 units kept after the cut, so the p-value is 0.10, the paper's number. This is the smallest p-value the design can produce: with  $J$  kept donors, no result can be rarer than one in  $J + 1$ . Per-period p-values (`p_value_by_period`) say in which years the gap stands out.

The predictor choice matters here. With all 30 lags, France's placebo fit is so tight (pre-treatment MSPE  $5.69e-04$ ) that a modest post-1990 gap gives it a larger ratio than Sweden, which then ranks second. The ratio statistic rewards fit as much as effect. Report the cut on badly fitted placebos and the predictor set together with the p-value, because both change the denominator and the ranking.

**When it breaks.** With few donors the test has little resolution. With donors whose own outcomes are hard to fit, the ratio statistic is noisy. And the test only ever speaks about the sharp null of no effect in any period; it does not give a confidence interval.

### 4.3 In-time placebos

**Procedure.** Move the treatment date to a year before the real one, use only the data before the real treatment, and fit the synthetic control as if the placebo date were real. A gap opening after the placebo date would mean the method finds effects where there are none.

```
pt_1980 <- synth_placebo(sc_sweden, type = "time", placebo_time = 1980)
pt_1970 <- synth_placebo(sc_sweden, type = "time", placebo_time = 1970)
c(placebo_1980 = pt_1980$estimate, placebo_1970 = pt_1970$estimate, real_1990 = sc_sweden$estimate)
#> placebo_1980 placebo_1970 real_1990
#> -0.04341465  0.07736121 -0.28367772

plot_synth(pt_1980, type = "path")
```

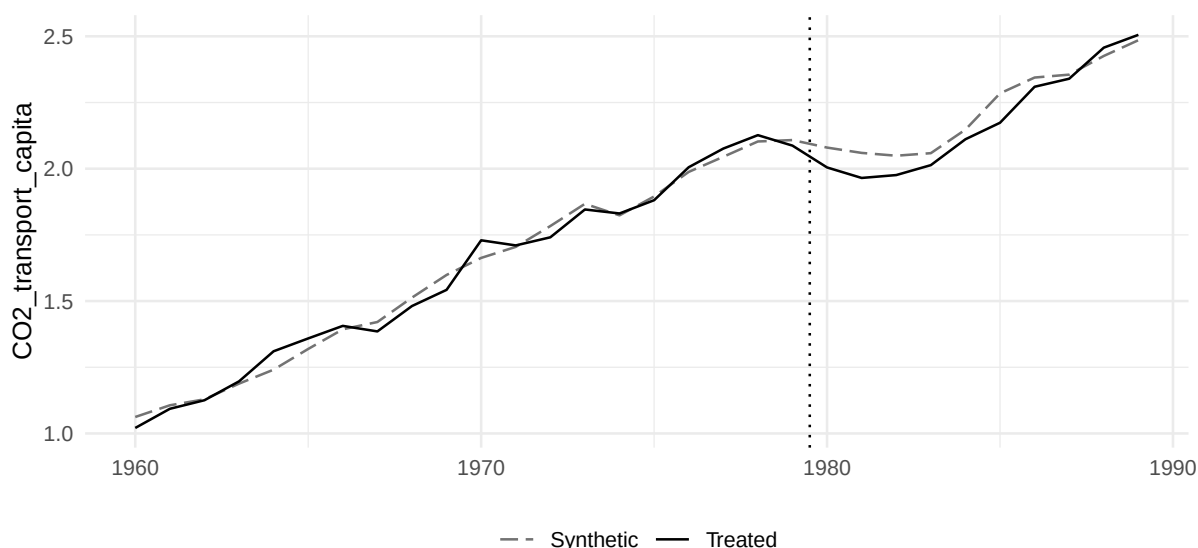


Figure 7: In-time placebo: treatment backdated to 1980, fitted on 1960 to 1979 and evaluated on 1980 to 1989.

The backdated gaps are small relative to the 1990 gap, which is what the design requires. Andersson uses fewer lags for these fits, because the placebo pre-period is shorter; the lecture keeps all lags for simplicity, and the replication follows the paper.

### 4.4 Conformal inference

**Procedure.** Chernozhukov, Wuthrich, and Zhu (2021) test a sharp null that fixes the effect in every post-treatment period,  $H_0 : \alpha_{0t} = \theta_t$ . Subtract  $\theta_t$  from the treated outcome after treatment, refit the synthetic control on all periods with every period's outcome as a predictor, and compute the residuals  $\hat{u}_t$ . Under the null the post-treatment residuals are like the others, so a statistic of the post-treatment residuals, here  $(T_1^{-1} \sum_{t>T_0} |\hat{u}_t|^q)^{1/q}$ , is compared with its value under all cyclic shifts of the residual vector (moving block permutations, which keep the dependence structure). The p-value is the share of shifts with a statistic at least as large. Inverting the test over a grid of constant effects gives a confidence set. Appendix 8.4 states the validity condition.

```
ci_sweden <- synth_conformal(sc_sweden, per_period = TRUE)
ci_sweden
#> Conformal inference (moving_block permutations, q = 1)
```



```
#> Average post-treatment effect -0.284; p-value for effect = 0: 0.391
#> 95% confidence set for a constant effect: [-0.681, 1.7]
```

```
ci_sweden$per_period
```

```
#>   time      gap    p_value  conf.low  conf.high
#> 1 1990 -0.06625895 0.22580645 -0.1608182 0.056668058
#> 2 1991 -0.14669560 0.03225806 -0.2412548 -0.052136364
#> 3 1992 -0.11356086 0.06451613 -0.2270319 0.009366153
#> 4 1993 -0.26340249 0.03225806 -0.3768736 -0.140475481
#> 5 1994 -0.26853576 0.03225806 -0.3725509 -0.126696900
#> 6 1995 -0.28435992 0.03225806 -0.3978310 -0.114153286
#> 7 1996 -0.35299250 0.03225806 -0.4664636 -0.173329947
#> 8 1997 -0.32896709 0.03225806 -0.4518941 -0.130392689
#> 9 1998 -0.33275383 0.03225806 -0.4462249 -0.115267578
#> 10 1999 -0.30265548 0.03225806 -0.4161266 -0.075713301
#> 11 2000 -0.30120259 0.03225806 -0.4241296 -0.036436717
#> 12 2001 -0.32838400 0.03225806 -0.4607669 -0.044706282
#> 13 2002 -0.32660391 0.03225806 -0.4589868 -0.052382118
#> 14 2003 -0.37820652 0.03225806 -0.5105895 -0.094528800
#> 15 2004 -0.38493198 0.03225806 -0.5078590 -0.101254260
#> 16 2005 -0.35933201 0.03225806 -0.4917149 -0.075654292
```

```
ggplot(ci_sweden$per_period, aes(x = time, y = gap)) +
  geom_hline(yintercept = 0, linetype = "dotted") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "grey80") +
  geom_line() + geom_point() +
  labs(x = NULL, y = "Gap with 95% conformal interval") + sc_theme()
```

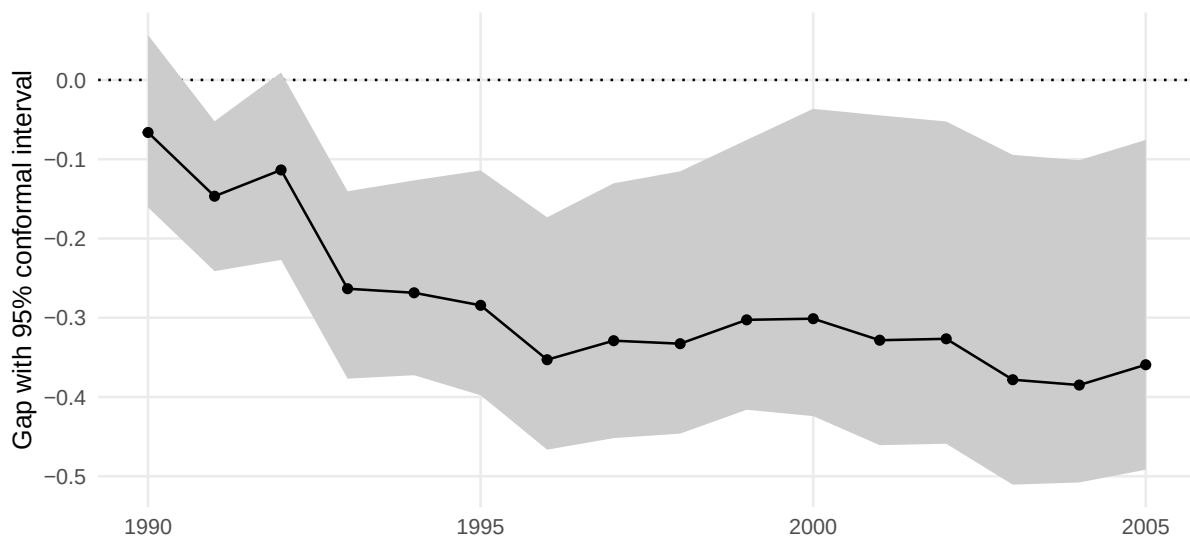


Figure 8: Per-period gaps for Sweden with 95 percent conformal intervals.

**Reading it.** The joint p-value for no effect in any post-treatment year is 0.39, and the confidence set for a constant effect is  $[-0.68, 1.70]$ . The joint test has little power for Sweden: the post-treatment block is 16 of 46 periods, so every cyclic shift overlaps it and the permutation distribution is contaminated by the effect itself. The per-period tests use one post-treatment period at a time and reject in every year from 1991 on, at the smallest p-value that 31 shifts allow. Conformal inference is the only tool here that produces an interval, and it needs only stationarity and weak dependence of the residuals, not a model for them.

**When it breaks.** The method is designed for a short post-treatment block relative to the pre-treatment period. With a long one, test period by period, or test the average effect over a few periods at a time. A structural break in the residuals after treatment violates the exchangeability the test relies on; the placebo test in space does not depend on it.

Table 9: Rejection rates at the ten percent level, 200 simulations per row. The specification test is a test of the design, not of the effect, and should not reject in either row.

| True effect | In-space placebo | Conformal | Specification test |
|-------------|------------------|-----------|--------------------|
| 0.0         | 0.095            | 0.065     | 0.080              |
| 1.5         | 0.445            | 0.255     | 0.085              |

## 4.5 Size and power of the three tests

The experiment draws panels from the factor model with 20 donors, 20 pre-treatment periods, and 5 post-treatment periods, with no effect and with an effect of 1.5 (about one noise standard deviation), and applies the three tests at the ten percent level.

```
inf_null <- run_inference_experiment(n_sims = n_sims, effect = 0, seed = 3)
inf_alt <- run_inference_experiment(n_sims = n_sims, effect = 1.5, seed = 4)
inf_tab <- summarise_inference(bind_rows(inf_null, inf_alt))
```

Both effect tests hold their size under the null (rejection rates 0.10 and 0.07). Against an effect of 1.5 the placebo test rejects in 44 percent of the samples and the conformal test in 26 percent, with five post-treatment periods and one treated unit. The specification test does not respond to the effect, as it should not: the factors have mean zero after treatment in this design, so the demeaned synthetic control and DiD estimate the same thing.

## 4.6 The specification test

**Procedure.** Under the factor model the demeaned synthetic control and DiD are both unbiased when the factors have mean zero after treatment, and both biased otherwise, with different biases. Ferman and Pinto (2021) contrast the two: fit the demeaned weights on all periods, form

$$\hat{u}_t = \hat{w}'(y_t - \bar{y}) - J^{-1}\iota'(y_t - \bar{y}),$$

the difference between the two counterfactuals (the treated outcome cancels), and compare the post-treatment mean of  $\hat{u}_t$  with its moving block permutations. A low p-value says the two estimators disagree by more than the pre-treatment variation explains.

```
st_sweden <- synth_spec_test(sc_dm)
st_sweden
#> Ferman-Pinto specification test: demeaned synthetic control versus DiD
#> Estimates: demeaned SC -0.291, DiD -0.214
#> p-value (mean contrast): 0.13; p-value (2-norm): 0.109

plot_synth(st_sweden)
```

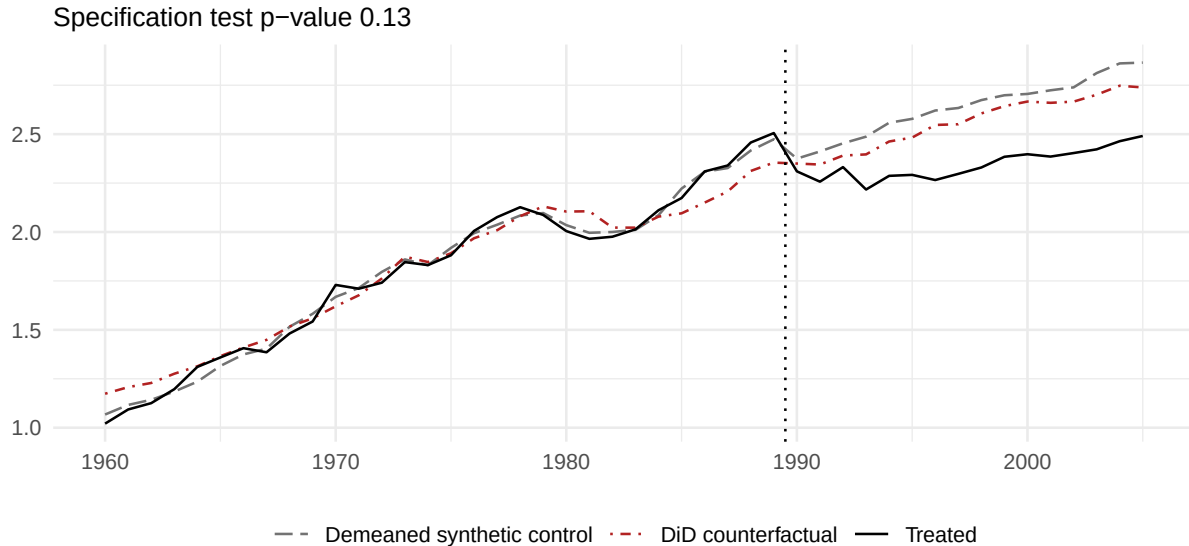


Figure 9: Sweden, its demeaned synthetic control, and the DiD counterfactual. The specification test asks whether the two counterfactuals diverge after 1990 by more than they wander before.

**Reading it.** For Sweden the test does not reject ( $p = 0.13$ ): the two counterfactuals move together, which supports the condition both need. Ferman and Pinto’s own illustration is the Basque Country (Abadie and Gardeazabal 2003), where the test rejects with  $p = 0.023$ ; the replication data reproduce it.

```
basque_path <- file.path(pkg_root, "data_raw", "09_Synthetic_Control", "ferman2021synthetic", "base_abadie.csv")
if (file.exists(basque_path)) {
  basque <- read.csv(basque_path)
  basque <- basque[!basque$regionname %in% c("Spain (Espana)", "Syntetic Basque Country"), ]
  basque$d <- as.integer(basque$regionname == "Basque Country (Pais Vasco)" & basque$year >= 1970)
  sc_basque <- synth_control(basque, id = "regionname", time = "year", y = "gdpcap", d = "d", demean = TRUE)
  synth_spec_test(sc_basque)
}

#> Ferman-Pinto specification test: demeaned synthetic control versus DiD
#> Estimates: demeaned SC -0.939, DiD -0.431
#> p-value (mean contrast): 0.0233; p-value (2-norm): 0.907
```

**How to read a rejection.** A rejection means at least one of the two estimators is biased. If the pre-treatment fit is close and the donors are many relative to the periods, the synthetic control may still be valid while DiD is not (Section 2.3, Step 1). If the fit is imperfect, both are suspect, and the demeaned synthetic control is the lesser evil because its weights lean toward the loadings. The test cannot tell which case applies; the fit and the concentration can.

## 4.7 Several treated units

With a handful of treated units adopting at the same time, the average mode fits one synthetic control to their average, and the placebo tools apply unchanged. With staggered adoption, or with treated units that differ enough that one synthetic control cannot serve all, the separate mode fits one synthetic control per unit from the donors that are still untreated over that unit’s window, and averages the gaps in event time. The jackknife over treated units gives a standard error for the average.

```
stag <- sim_synth_panel(n_donors = 25, n_treated = 5, t_pre = 15, t_post = 8, effect = c(0.5, 1, 1.5, 2),
  adoption = c(14, 15, 17, 19, 21), seed = 7)
sc_stag <- synth_control(stag, id = "id", time = "time", y = "y", d = "d",
  treated_units = "separate", horizon = 4, demean = TRUE)

sc_stag
#> Demeaned synthetic control fitted separately to 5 treated unit(s)
#> Average post-treatment effect: 1.368 (jackknife SE over units 0.178)
#> Per-unit estimates in `unit_table`, event-time gaps in `event_gap`, fits in `units`
```

```
sc_stag$unit_table
#>      id adoption n_donors pre_rmspe estimate
#> 26 26      14      27 0.8409041 1.3468694
#> 27 27      15      27 0.6692112 0.7088365
#> 28 28      17      26 0.7563422 1.4196076
#> 29 29      19      25 1.0063695 1.6798826
#> 30 30      21      25 1.4811078 1.6844839
```

```
plot_synth(sc_stag)
```

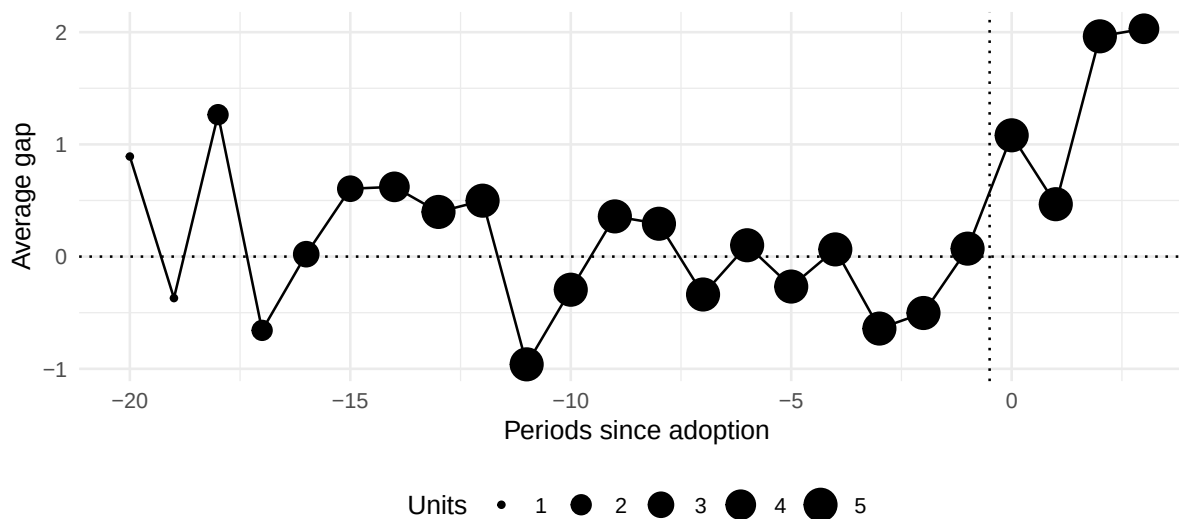


Figure 10: Average gap by event time over five treated units with staggered adoption; the true effect rises from 0.5 to 2 over the first four periods.

Each unit's donor pool shrinks as adoption spreads, and units treated late have fewer donors. Two refinements are in the literature and not in the package: pooled placebo p-values across treated units (Cavallo et al. 2013) and partially pooled weights that trade unit-level fit for fit of the average (Ben-Michael, Feller, and Rothstein 2022). Synthetic DiD with cohort-by-cohort weights (Section 07) is the other option.

## 5 Robustness and design checks

### 5.1 Leave-one-out

Drop each donor with positive weight in turn and refit. The spread of the refitted estimates measures how much the answer rests on any single donor. Andersson (2019, Figure 9) runs this for the six donors with weight above 0.001.

```
loo_sweden <- synth_loo(sc_and)
loo_sweden
#> Leave-one-donor-out: full estimate -0.297; range over refits [-0.352, -0.259]
#>      dropped weight estimate pre_rmspe
#> Belgium 0.1667 -0.286 0.0366
#> Denmark 0.3866 -0.352 0.0507
#> Greece 0.1163 -0.270 0.0369
#> New Zealand 0.2166 -0.290 0.0360
#> Switzerland 0.0270 -0.302 0.0349
#> United States 0.0869 -0.259 0.0446
```

```
plot_synth(loo_sweden)
```

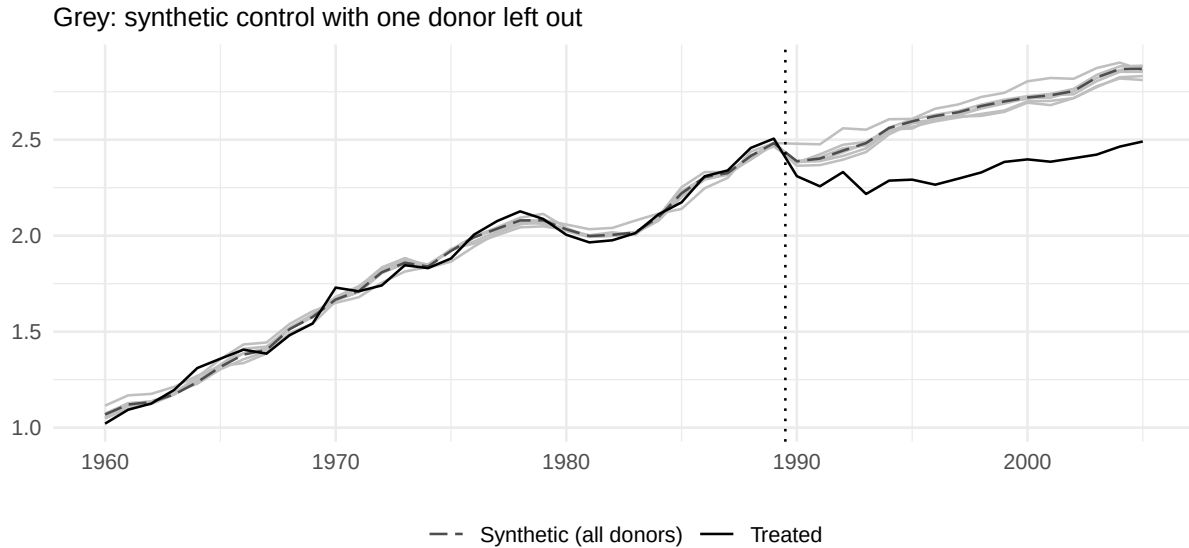


Figure 11: Synthetic Sweden with each positive-weight donor left out in turn (grey), against the full synthetic control and Sweden.

The estimate is robust to any single donor; leaving out Denmark, the largest donor, gives a somewhat larger reduction, as in the paper. A leave-one-out spread comparable to the estimate would say the result is one donor's story.

## 5.2 The donor pool

The donor pool is a modelling decision, and it is made before any weight is estimated. Four rules.

- **Unaffected.** No donor should be exposed to the treatment through spillovers (fuel tourism across a border, trade diversion).
- **No policy of its own.** Donors with a similar policy in the window are removed; Andersson drops Finland, Norway, and the Netherlands, which introduced carbon taxes in the 1990s.
- **Comparable.** Donors should share the structure that drives the outcome, so that the factor loadings of the treated unit are inside their span.
- **Complete.** A balanced panel over the whole window.

A useful check is to restrict the pool to the units one would defend as comparable and see whether the estimate moves.

```
europe <- c("Belgium", "Denmark", "France", "Greece", "Iceland", "Poland", "Portugal", "Spain", "Switzerland", "Sweden")
sc_europe <- synth_control(carbon[carbon$country %in% europe, ], "country", "year", "CO2_transport_capita", "treated",
  x = xs, lags = c(1989, 1980, 1970), pre_window = 1980:1989, v = "mspe",
  v_control = list(maxit = 2000, starts = 3))
c(all_donors = sc_and$estimate, european_donors = sc_europe$estimate,
  pre_rmspe_all = sc_and$pre_rmspe, pre_rmspe_europe = sc_europe$pre_rmspe)
#>      all_donors  european_donors  pre_rmspe_all pre_rmspe_europe
#>      -0.29654392    0.08091787    0.03471442    0.15927832
```

Restricting the pool to European donors removes New Zealand and the United States, which carry 0.30 of the weight, and the pre-treatment fit deteriorates by a factor of 4.6. The estimate that comes with that fit changes sign. It is not an alternative estimate; it is a synthetic control that failed, and the right reading is that Sweden's emissions path has no counterpart among European donors alone. A donor-pool restriction is informative only when the fit survives it.

### 5.3 Confounders that move with the treatment

Sweden entered a deep recession in 1991 to 1993, right after the tax. If the recession drove the emissions gap, the gap should close when the economy recovers. Andersson's check applies the same donor weights to GDP per capita: if synthetic Sweden tracks real Sweden's GDP through the recession, the recession is not what separates the emission paths. `synth_control(weights = )` reuses the weights on another outcome.

```
# GDP per capita is missing for Poland before 1970; the check uses 1970 to 2005.
gdp_sweden <- synth_control(carbon[carbon$year >= 1970, ], "country", "year", "GDP_per_capita", "treated", weights = sc_and)
gaps <- merge(data.frame(year = sc_and$effects$time, emissions = sc_and$effects$gap),
             data.frame(year = gdp_sweden$effects$time, gdp = gdp_sweden$effects$gap), by = "year")
change <- function(v, from, to) v[gaps$year == to] - v[gaps$year == from]
recession_tab <- data.frame(
  recession = c("1975 to 1978", "1989 to 1993"),
  gdp_gap_change = c(change(gaps$gdp, 1975, 1978), change(gaps$gdp, 1989, 1993)),
  emissions_gap_change = c(change(gaps$emissions, 1975, 1978), change(gaps$emissions, 1989, 1993))
)
recession_tab
#>      recession gdp_gap_change emissions_gap_change
#> 1 1975 to 1978      -2314.223         0.08693153
#> 2 1989 to 1993      -2304.758        -0.28876729

recessions <- data.frame(start = c(1976, 1991), end = c(1978, 1993))
ggplot() +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf), fill = "grey85") +
  geom_hline(yintercept = 0, linetype = "dotted") + geom_vline(xintercept = 1989.5, linetype = "dotted") +
  geom_line(data = gaps, aes(x = year, y = emissions, colour = "CO2 emissions gap")) +
  geom_line(data = gaps, aes(x = year, y = gdp / 5000, colour = "GDP per capita gap")) +
  scale_y_continuous(sec.axis = sec_axis(~ . * 5, name = "GDP gap, thousand dollars")) +
  scale_colour_manual(values = c("CO2 emissions gap" = "black", "GDP per capita gap" = "#B22222")) +
  labs(x = NULL, y = "Emissions gap, tons per capita", colour = NULL) + sc_theme()
```

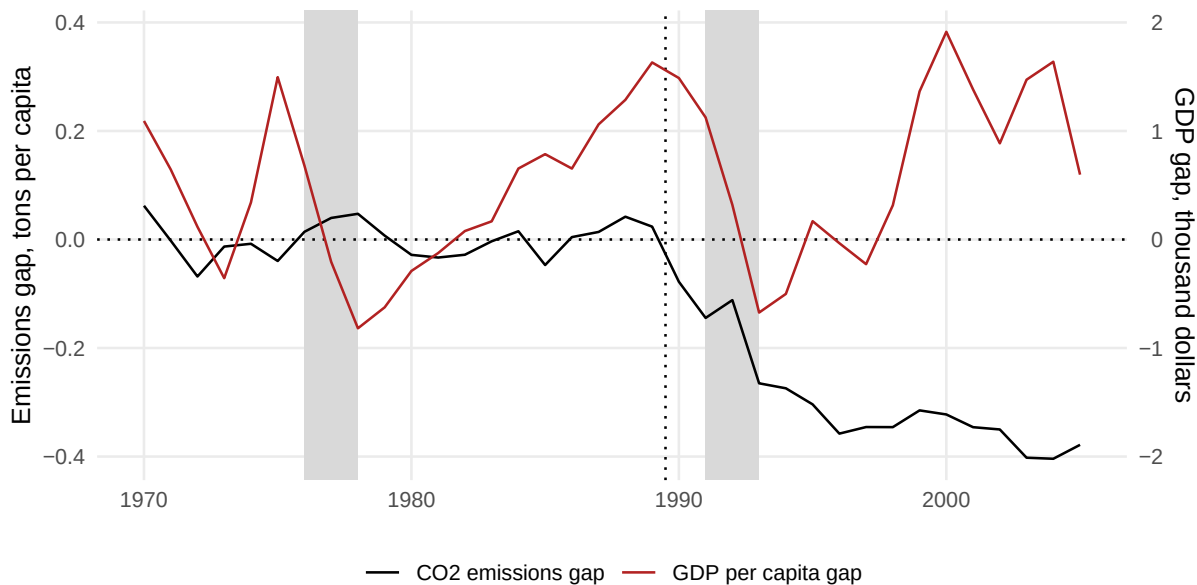


Figure 12: Gaps between Sweden and synthetic Sweden in CO2 emissions (left axis) and in GDP per capita (right axis, thousands of 2005 dollars), with the two recessions shaded.

The GDP gap falls by a similar amount in both recessions. The emissions gap rises slightly in the first and falls in the second, and it does not close when GDP recovers after 1993. The recession is not the explanation. The same logic applies to any confounder that can be observed: build its synthetic version with the same weights and ask whether it moved.

## 5.4 Justifying the design: the Ferman-Pinto checklist

Ferman and Pinto (2021, Section 6) turn their results into six requests to the applied researcher. Each maps to an output of the package.

| Request                                                                                                                                                          | Where to look                                       | Sweden                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------|
| 1. Argue that shocks specific to the treated unit are unrelated to treatment; confounders that also move the donors are allowed                                  | the design, in words                                | the tax followed a political process, not an emissions shock |
| 2. Show the pre-treatment fit after removing common trends, not only in levels                                                                                   | <code>plot_synth(type = "demeaned_path")</code>     | Figure 4: close                                              |
| 3. Keep the donors few relative to the pre-treatment periods, or a good fit may be overfitting                                                                   | <code>glance(): n_donors, t_pre</code>              | 14 donors, 30 years                                          |
| 4. Report the weight concentration $\ w\ ^2$ ; diluted weights are the case where the estimator is asymptotically unbiased even with imperfect fit (Ferman 2021) | <code>concentration</code>                          | 0.25, about 4.1 effective donors                             |
| 5. Compare the demeaned synthetic control with DiD; a large difference says both are biased                                                                      | <code>synth_spec_test()</code>                      | $p = 0.13$ , no rejection                                    |
| 6. Use the original synthetic control only if it recovers the treated unit's level; otherwise demean or abandon                                                  | <code>balance, intercept</code> of the demeaned fit | intercept 0.057: levels match                                |

Sweden passes each request, which is why the paper's estimate survives a theory published two years after it. The checklist is also a map of the ways an application fails: a fit that is good only in levels, a donor pool larger than the pre-period, weights on one or two donors, or a demeaned estimate far from DiD.

## 6 Practice

### 6.1 Replications and anchors

| Paper                                                      | What it anchors                                                                                                                  | Status                                                  |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Andersson (2019), carbon taxes and CO2 emissions in Sweden | the full workflow: predictors and V, weights, balance, path and gap, in-time and in-space placebos, leave-one-out, the GDP check | <code>inst/replications/09_Synthetic_Control/and</code> |
| Ferman and Pinto (2021), imperfect pretreatment fit        | the theory of Section 2, the demeaned estimator, the specification test on the Basque Country, the Monte Carlo of Section 4      | setup described below; notebook to be written           |

| Paper                                                                 | What it anchors                                                         | Status                                                                                |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Abadie, Diamond, and Hainmueller (2010), California's Proposition 99  | the canonical application and the placebo test in space                 | data in the <code>synthdid</code> package; see lecture 07                             |
| Berman and Israeli (2022), descriptive analytics and online retailers | staggered adoption in a marketing panel, cohort-by-cohort synthetic DiD | lecture 07; <code>treated_units = "separate"</code> is the synthetic control analogue |

**Ferman and Pinto (2021): setup and construction.** The replication folder holds the authors' code and data. Two parts.

1. *Basque application.* Data: real GDP per capita for 17 Spanish regions, 1955 to 1997, from Abadie and Gardeazabal (2003); the Basque Country is treated from 1970 (terrorism), so  $T_0 = 15$  and  $J = 16$  after dropping Spain and the authors' pre-built synthetic unit. Construction: (i) the original synthetic control with all 15 outcome lags, solved on raw outcomes (`synth_control()` with defaults); (ii) the demeaned version (`demean = TRUE`); (iii) the specification of Abadie and Gardeazabal with covariates, for comparison; (iv) plots in levels and after subtracting each period's regional mean (`plot_synth(type = "demeaned_path")`); (v) the specification test (`synth_spec_test()`), which gives  $p = 0.023$ , and the weights table (Appendix A.2 of the paper).
2. *Monte Carlo (their Table 1).* Data: monthly CPS employment rates for the 50 states and DC, 1982 to 2019. Construction: estimate a linear factor model by interactive fixed effects (Bai 2009) with the number of factors chosen by the Bai-Ng criterion (the authors find four), fit ARMA processes to each factor and each state's residuals, and simulate with  $T_1 = 12$  and  $T_0 \in \{120, 240, 480, 1200\}$ . Panels A and B treat the state with the second largest and second smallest fixed effect; Panels C and D the state with the second largest and smallest loading on the first factor, adding a break of two standard deviations to that factor after treatment. Compare the original synthetic control, the demeaned synthetic control, and DiD on bias and standard error. `sim_synth_panel()` is the stylized version of this design; the authors' version needs an interactive fixed effects estimator (the `gsynth` package), which is not part of this package.

## 6.2 A synthetic control checklist

1. Name the treated unit, the treatment date, and the window. Argue no anticipation (or date the treatment at the announcement) and no interference with the donors.
2. Describe the donor pool and every exclusion, with reasons.
3. State the predictors, the window over which covariates are averaged, and how  $V$  was chosen. Report the balance table.
4. Report the weights, the concentration  $\|w\|^2$ , and the effective number of donors.
5. Show the path plot and the demeaned path plot. Report the pre-treatment RMSPE and the ratio of donors to pre-treatment periods.
6. Report the average and the per-period gaps.
7. Run in-space placebos; report the ratio ranking, the cut rule, and the p-value. Run at least one in-time placebo.
8. Report a conformal interval, per period when the post-treatment block is long.
9. Report the demeaned estimate next to the original and the specification test against DiD.
10. Report leave-one-out, at least one donor-pool restriction, and the same-weights check on the confounders a referee will name.

## 7 Extensions

- **Synthetic DiD** (Arkhangelsky et al. 2021) adds period weights and two-way effects; Section 07 and `sdid_weights()`. It is the demeaned synthetic control with a data-driven choice of which pre-treatment periods matter.



- **Penalized synthetic control** (Abadie and L’Hour 2021) adds a penalty on the distance between the treated unit and each donor it uses. The weights become unique and sparse, and the method extends to many treated units in disaggregated data.
- **Staggered adoption with partial pooling** (Ben-Michael, Feller, and Rothstein 2022) fits one synthetic control per unit and one for the average, and trades off their fits. The separate mode of `synth_control()` is the unpooled corner.
- **Interactive fixed effects and generalized synthetic control** (Bai 2009; Xu 2017) estimate the factors and loadings directly and impute the counterfactual from them. They need many donors and periods, and they extrapolate; `gsynth` implements them.
- **Matrix completion** (Athey et al. 2021) treats the untreated outcomes as a matrix with missing entries and fills them under a nuclear-norm penalty; synthetic control and regression on lags are the two one-sided special cases.
- **Prediction intervals** (Cattaneo, Feng, and Titiunik 2021) separate the uncertainty from the weights and from the post-treatment shocks and give intervals under explicit models for each; `scpi` implements them.
- **Bayesian structural time series** (Brodersen et al. 2015) models the treated unit’s series with the donors as regressors and reports a posterior for the counterfactual; `CausalImpact` is the common choice in marketing.
- **Geo experiments** (GeoLift) use the augmented synthetic control to evaluate market-level marketing tests, where a few regions receive a campaign and the rest are donors. The randomization of the treated regions restores design-based inference.
- **Doubly robust panel identification** (Arkhangelsky and Imbens 2022) puts an assignment model and an outcome model on the same footing and shows when synthetic control, DiD, and their combinations are valid.

## 8 Appendix

### 8.1 The bias bound under perfect fit

Setting: the factor model of Section 2.2 with  $F$  factors,  $\Lambda = (\lambda_1, \dots, \lambda_{T_0})'$  the  $T_0 \times F$  matrix of pre-treatment factors, and weights  $w^*$  with  $c'w^* = c_0$  and  $\mu'w^* = \mu_0$ . Suppose the fitted weights  $\hat{w}$  give an exact pre-treatment fit:  $Y_{0s} = \sum_j \hat{w}_j Y_{js}$  for all  $s \leq T_0$ .

*Step 1.* By the decomposition of Section 2.3, an exact fit means that for every  $s \leq T_0$ ,

$$(c_0 - c'\hat{w}) + \lambda'_s(\mu_0 - \mu'\hat{w}) = \varepsilon'_s \hat{w} - \varepsilon_{0s}.$$

Stack the  $T_0$  equations. With  $\tilde{\Lambda} = (\iota, \Lambda)$  of full column rank, the level and loading mismatches are the least-squares projection of the noise contrast  $e = (\varepsilon'_s \hat{w} - \varepsilon_{0s})_{s \leq T_0}$  on the factor span:

$$\begin{pmatrix} c_0 - c'\hat{w} \\ \mu_0 - \mu'\hat{w} \end{pmatrix} = (\tilde{\Lambda}'\tilde{\Lambda})^{-1}\tilde{\Lambda}'e.$$

*Step 2.* The post-treatment bias is  $\lambda'_t(\mu_0 - \mu'\hat{w})$  plus the level term, a linear function of the projected noise. Its mean is bounded by a constant times the expected absolute size of  $(\tilde{\Lambda}'\tilde{\Lambda}/T_0)^{-1}(\tilde{\Lambda}'e/T_0)$ . When  $\tilde{\Lambda}'\tilde{\Lambda}/T_0$  converges to a positive definite matrix and the noise has bounded moments,  $\tilde{\Lambda}'e/T_0$  is an average of  $T_0$  mean-zero terms with variance of order  $\sigma^2(1 + \|\hat{w}\|^2)/T_0$ . The bound therefore shrinks in  $T_0$  and grows with the noise variance and, through  $\|\hat{w}\|^2 \leq 1$  and the sum over donors, with  $J$ . Abadie, Diamond, and Hainmueller (2010, Theorem 1) give the explicit constant. The conclusion used in the text: under an exact fit the bias vanishes as  $T_0$  grows, at a rate governed by the noise relative to the factors.

The step that fails with an imperfect fit is the first one. When the fit is not exact, the stacked equations hold only up to the fitting error, and that error is chosen by the estimator to trade off against the noise term, which is Appendix 8.2.

## 8.2 The limit objective with imperfect fit

Setting: all pre-treatment outcomes as predictors,  $J$  fixed,  $T_0 \rightarrow \infty$ . Ferman and Pinto's Assumption 4: pre-treatment averages of  $\lambda_s$ ,  $\varepsilon_s$ , and their cross products converge in probability, with  $T_0^{-1} \sum_s \lambda_s \rightarrow 0$ ,  $T_0^{-1} \sum_s \lambda_s \lambda'_s \rightarrow \Omega_0$ ,  $T_0^{-1} \sum_s \varepsilon_s \varepsilon'_s \rightarrow \sigma^2 I_{J+1}$ , and cross terms to zero. (Normalizing the factor mean to zero is without loss: a nonzero mean is absorbed in  $c_j$ .)

*Derivation of  $Q_0$ .* The objective is

$$Q_{T_0}(w) = \frac{1}{T_0} \sum_{s \leq T_0} [(c_0 - c'w) + \lambda'_s(\mu_0 - \mu'w) + (\varepsilon_{0s} - \varepsilon'_s w)]^2.$$

Expand the square. The squared level term is constant in  $s$ . The squared factor term averages to  $(\mu_0 - \mu'w)' \Omega_0 (\mu_0 - \mu'w)$ . The squared noise term averages to  $\sigma^2(1 + w'w)$ , because  $\varepsilon_{0s}$  and  $\varepsilon'_s w$  are uncorrelated with variance  $\sigma^2$  and  $\sigma^2 w'w$ . Every cross term averages to zero. Hence  $Q_{T_0}(w) \rightarrow_p Q_0(w)$  pointwise, and uniformly on the simplex because  $Q_{T_0}$  is quadratic in  $w$  with converging coefficients. Continuity and compactness give  $\hat{w} \rightarrow_p \bar{w} = \arg \min_{\Delta} Q_0$  when the minimizer is unique (Proposition 1).

*Why  $\bar{w}$  misses the loadings.* Let  $\mathcal{W}^* = \{w \in \Delta : c'w = c_0, \mu'w = \mu_0\}$ . On  $\mathcal{W}^*$  the first two terms vanish and  $Q_0(w) = \sigma^2(1 + w'w)$ . The minimizer of  $Q_0$  over all of  $\Delta$  equals a point of  $\mathcal{W}^*$  only if that point also minimizes  $w'w$  over  $\Delta$  up to the trade-off with the first two terms, which happens only when  $\sigma^2 = 0$  or by coincidence. Otherwise  $\bar{w}$  lies outside  $\mathcal{W}^*$ , and the mismatch  $\mu_0 - \mu'\bar{w}$  is a fixed nonzero vector.

*The demeaned estimator (Proposition 2).* Replace every series by its deviation from its own pre-treatment mean. The level term drops from the objective, so the limit is  $\tilde{Q}_0(w) = (\mu_0 - \mu'w)' \Omega_0 (\mu_0 - \mu'w) + \sigma^2(1 + w'w)$  and the estimator converges to  $\alpha_{0t} + \lambda'_t(\mu_0 - \mu'\tilde{w}) + (\varepsilon_{0t} - \varepsilon'_t \tilde{w})$ . It is asymptotically unbiased when  $\mathbb{E}[\lambda_t] = 0$  for  $t > T_0$ , whatever  $\tilde{w}$  is.

*Efficiency against DiD (Proposition 3).* Assume the factor and noise moments after treatment equal the pre-treatment limits. Then for any fixed  $w$  the asymptotic mean squared error of  $\hat{\alpha}_{0t}(w)$  is  $\tilde{Q}_0(w)$ : the squared factor mismatch plus the noise variance. The demeaned weights minimize  $\tilde{Q}_0$  over the simplex, and DiD uses  $w = \iota/J$ , which is in the simplex. Hence  $\tilde{Q}_0(\tilde{w}) \leq \tilde{Q}_0(\iota/J)$ : the demeaned synthetic control weakly dominates DiD in asymptotic mean squared error. The inequality is strict unless uniform weights already minimize the trade-off.

## 8.3 One family: weights plus an outcome model

Write every estimator as  $\hat{Y}_{0t}^N = \hat{m}_t(Z_0) + \sum_j w_j(Y_{jt} - \hat{m}_t(Z_j))$ .

- *Synthetic control:*  $\hat{m}_t \equiv 0$ .
- *Demeaned synthetic control:*  $\hat{m}_t(Z_j) = \bar{Y}_j$ , the unit's pre-treatment mean. Then  $\hat{Y}_{0t}^N = \bar{Y}_0 + \sum_j w_j(Y_{jt} - \bar{Y}_j)$ , which is the intercept form  $\hat{a} + \sum_j w_j Y_{jt}$  with  $\hat{a} = \bar{Y}_0 - \sum_j w_j \bar{Y}_j$ . Minimizing the pre-treatment fit over  $(a, w)$  and over  $w$  with the series demeaned give the same  $w$ , because the optimal  $a$  for given  $w$  is exactly that expression.
- *Augmented synthetic control:*  $\hat{m}_t(Z) = \hat{a}_t + \hat{\beta}'_t Z$  from a ridge regression of  $Y_{jt}$  on the donors' predictors. Substituting gives  $\hat{Y}_{0t}^N = \sum_j w_j Y_{jt} + \hat{\beta}'_t (Z_0 - \sum_j w_j Z_j)$ : the synthetic control plus the outcome model applied to the imbalance.
- *Synthetic DiD:*  $\hat{m}_t(Z_j) = \hat{a}_j + \hat{b}_t$ , two-way effects estimated with period weights  $\lambda_t$  on the pre-treatment periods; the unit weights are then fitted to the residuals.
- *DiD:* the same outcome model with uniform unit and period weights.

The error decomposition of Section 2.4 follows by substituting  $Y_{jt} = m_t(Z_j) + e_{jt}$ .

## 8.4 Validity of the moving-block conformal test

Under the sharp null the adjusted treated outcome has no effect in any period, and the refit on all  $T$  periods treats every period alike. Let  $u_t$  be the population residual of that fit and  $\hat{u}_t$  its estimate. Chernozhukov,

Wuthrich, and Zhu (2021) assume that  $(u_t)$  is stationary and weakly dependent, and that the estimator is consistent so that  $\hat{u}_t - u_t$  is small uniformly in  $t$ . For a statistic  $S(\hat{u}_\pi)$  of a block of  $T_1$  consecutive residuals, the set of  $T$  cyclic shifts  $\pi$  treats every block of  $T_1$  consecutive periods as the post-treatment block in turn. Under stationarity each block has the same distribution, so the rank of the observed block among the  $T$  blocks is approximately uniform, and  $\mathbb{P}(\hat{p} \leq a) \rightarrow a$  as  $T \rightarrow \infty$  with  $T_1$  fixed. With independent residuals the test is exact in finite samples. The approximation degrades when  $T_1/T$  is not small, because the blocks overlap the true post-treatment period heavily and the effect, if the null is false, contaminates most of them; this is the low power seen for Sweden. The specification test of Ferman and Pinto applies the same argument to the contrast  $\hat{u}_t$  of Section 4.6, with the treated outcome cancelled.

## 8.5 The nested problem for the importance weights

With predictors  $Z_0$  (a vector of length  $K$ ) and  $Z$  ( $K \times J$ ), standardized by their standard deviation across units, the inner problem for given  $V = \text{diag}(v_1, \dots, v_K)$  is the quadratic program

$$w(V) = \arg \min_{w \in \Delta} (Z_0 - Zw)'V(Z_0 - Zw),$$

solved exactly by an active-set method (**quadprog**) or by Frank-Wolfe steps on the simplex. The outer problem chooses  $V$  in the simplex to minimize the pre-treatment mean squared prediction error of the outcome,  $T_0^{-1} \sum_{s \leq T_0} (Y_{0s} - Y'_s w(V))^2$ . It is not convex and has flat regions: many  $V$  give nearly the same  $w$ . `synth_control(v = "mspe")` runs Nelder-Mead from the equal-weight and regression-based starts, refines by BFGS, and keeps the best. When every pre-treatment outcome is a predictor, the outer objective is already the inner objective restricted to the lag rows, so putting all of  $V$  on the lags attains the outer minimum and covariates receive zero weight (Kaul et al. 2022).

## 8.6 References

- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association*.
- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2015. "Comparative Politics and the Synthetic Control Method." *American Journal of Political Science*.
- Abadie, Alberto, and Javier Gardeazabal. 2003. "The Economic Costs of Conflict: A Case Study of the Basque Country." *American Economic Review*.
- Abadie, Alberto, and Jeremy L'Hour. 2021. "A Penalized Synthetic Control Estimator for Disaggregated Data." *Journal of the American Statistical Association*.
- Andersson, Julius J. 2019. "Carbon Taxes and CO2 Emissions: Sweden as a Case Study." *American Economic Journal: Economic Policy*.
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review*.
- Arkhangelsky, Dmitry, and Guido W. Imbens. 2022. "Doubly Robust Identification for Causal Panel Data Models." *The Econometrics Journal*.
- Athey, Susan, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. 2021. "Matrix Completion Methods for Causal Panel Data Models." *Journal of the American Statistical Association*.
- Bai, Jushan. 2009. "Panel Data Models with Interactive Fixed Effects." *Econometrica*.
- Ben-Michael, Eli, Avi Feller, and Jesse Rothstein. 2021. "The Augmented Synthetic Control Method." *Journal of the American Statistical Association*.
- Ben-Michael, Eli, Avi Feller, and Jesse Rothstein. 2022. "Synthetic Controls with Staggered Adoption." *Journal of the Royal Statistical Society: Series B*.
- Berman, Ron, and Ayelet Israeli. 2022. "The Value of Descriptive Analytics: Evidence from Online Retailers." *Marketing Science*.
- Brodersen, Kay H., Fabian Gallusser, Jim Koehler, Nicolas Remy, and Steven L. Scott. 2015. "Inferring Causal Impact Using Bayesian Structural Time-Series Models." *Annals of Applied Statistics*.

- Cattaneo, Matias D., Yingjie Feng, and Rocio Titiunik. 2021. "Prediction Intervals for Synthetic Control Methods." *Journal of the American Statistical Association*.
- Cavallo, Eduardo, Sebastian Galiani, Ilan Noy, and Juan Pantano. 2013. "Catastrophic Natural Disasters and Economic Growth." *Review of Economics and Statistics*.
- Chernozhukov, Victor, Kaspar Wuthrich, and Yinchu Zhu. 2021. "An Exact and Robust Conformal Inference Method for Counterfactual and Synthetic Controls." *Journal of the American Statistical Association*.
- Doudchenko, Nikolay, and Guido W. Imbens. 2016. "Balancing, Regression, Difference-in-Differences and Synthetic Control Methods: A Synthesis." NBER Working Paper 22791.
- Ferman, Bruno. 2021. "On the Properties of the Synthetic Control Estimator with Many Periods and Many Controls." *Journal of the American Statistical Association*.
- Ferman, Bruno, and Cristine Pinto. 2021. "Synthetic Controls with Imperfect Pretreatment Fit." *Quantitative Economics*.
- Hsiao, Cheng, H. Steve Ching, and Shui Ki Wan. 2012. "A Panel Data Approach for Program Evaluation: Measuring the Benefits of Political and Economic Integration of Hong Kong with Mainland China." *Journal of Applied Econometrics*.
- Kaul, Ashok, Stefan Klossner, Gregor Pfeifer, and Manuel Schieler. 2022. "Standard Synthetic Control Methods: The Case of Using All Preintervention Outcomes Together with Covariates." *Journal of Business and Economic Statistics*.
- Xu, Yiqing. 2017. "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models." *Political Analysis*.